Lesson 5 Quantifiers

1. For each statement, determine if it is a universal or existence statement. If it is a universal statement, write it in the form, "For all …, if … then …"
   a) The product of two odd integers is odd.
   b) Some matrices are not invertible.
   c) A differentiable function is continuous.
   d) There are continuous functions which are not differentiable.
   e) No even integers are prime.
   f) Consecutive integers do not have the same parity.

2. Write the following statements in English.
   a) $\forall x \in \mathbb{R}, \exists y \in \mathbb{N}, y > x$
   b) $\exists y \in \mathbb{N}, \forall x \in \mathbb{R}, y > x$
   c) $\exists x, y \in \mathbb{N}, x + y = 0$
   d) $\forall x, y \in \mathbb{N}, xy > 0$

3. Write a negation for each of the following statements.
   a) For every integer n, either n is even or $n + 1$ is even.
   b) An invertible matrix has nonzero determinant.
   c) No even integers are prime.
   d) Every set is a subset of itself.
   e) There exists a continuous function which is not differentiable.
   f) There exists a real number x such that for all real numbers y, $xy = 1$.
   g) For every real number x, there exists a real number y such that $xy = 1$.
   h) For all non-zero integers x and y, $x + y \neq x - y$.
   i) For all $a, b \in \mathbb{R}$, if $a > 0$, then there is $n \in \mathbb{N}$ such that $na > b$.
   j) If $a, b \in \mathbb{R}$ and $a < b$, then there is $r \in \mathbb{Q}$ such that $a < r < b$.

4. Consider the following definition.
   - A subset S of $\mathbb{R}$ is ***bounded below*** if there is a real number L such that $x \geq L$ for all $x \in S$. L is called a ***lower bound*** of S.

   If L is a lower bound, then all numbers lower than L are lower bounds. The following sets are bounded below, find the ***greatest*** ***lower bound*** (***glb***).
   a) $\mathbb{N}$
   b) $S = \{x \in \mathbb{Q} : x \geq -1\}$
   c) $S = \{x \in \mathbb{Q} : |x| < 2\}$
   d) $S = (2, 10) \cup [3, \infty)$
   e) $S = (2, 10) \cap [3, \infty)$

5. Consider the following definition.
   - A set S is ***closed*** under an operation $*$ if $x * y \in S$ for all $x, y \in S$.

   a) Prove or disprove: $\mathbb{Z}$ is closed under addition.
   b) Prove or disprove: $\mathbb{N}$ is closed under division.